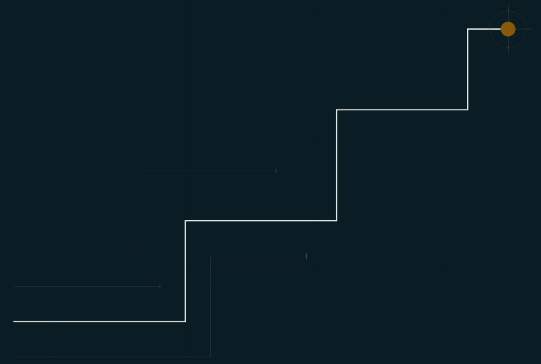
LUMEN BANK · DISPUTE OPERATIONS

From complaint to cited draft.

A copilot that does the investigation, cites every fact, and stops at the human.



FINAL DEMO AND TECHNICAL DEFENSE · 29 OCTOBER 2026 · KANIT, MANI, ZACH

Fictional training data. Mock system of record. No customer data.

Every complaint restarts the same investigation.

Every case sits in the queue while someone rebuilds the same five sources by hand.

POLICY BY DATE
Version 1 or version 2, chosen from the reported date, not the date of the call.

FACTS FROM TWO SYSTEMS
The merchant record and the transaction record are two stores with two shapes.

THE CLOCK
A deadline is counted from the reported date, and a day count can be wrong by one.

“I found a card purchase at Morrow Supplies for 117 crowns. I do not recognize it and need help checking it.”

CARD-D01-INTAKE · UNTRUSTED TEXT

One contract, six citations, one gate.

The object's shape is fixed before any prose exists.

CONTRACT · CARD-D01 · 10 KEYS · 7 SHOWN

case_id	CARD-D01
classification	unauthorized
routing	ready_for_review
facts	TX-D01 · 117 XCR · 2026-10-01
policy_ids	CARD-POLICY-V2 · CARD-PROCEDURE-01
deadline	2026-10-31
citations	6 claims from 5 sources

```
citation[0]  claim customer_evidence  source intake
citation[3]  claim transaction_record  source transactions
```

ONE GATE

sha256 of the draft

af65f686...ebd4

A human approves

201 pending review

One record or none

changed draft: refused, no write

409 duplicate_case

A stop is an outcome, not an error: the contract prints the reason and no record is written.

Five complaints. Five ways it can go wrong.

Each row is a case you can run in the terminal. Every row now passes on main.

CASE	WHAT IT SHOULD DO	WHAT YOU SEE	STATUS
D01	Record from the draft.	Hash bound, approved, record written: 201, then a 200 replay. A changed draft stops before any write; 409 on a second record.	ON MAIN
D05	Decide a contradiction.	Both contradictory sources cited; route manual_review, no deadline. A decided outcome, not a guess.	ON MAIN
D08	Refuse an instruction.	The instruction sentence is filtered, never read as the customer's account.	ON MAIN
D10	Retry a fault.	One injected transaction failure. The ladder retries once: 3 attempts, 1 retry, 1.3 s.	ON MAIN
D09	Refuse a redelivery.	Duplicate of D02: TX-D02, 134 XCR, 16 Nov. The excerpt join runs on main.	ON MAIN

ON MAIN = on 8553bb3 today: every row is a passing dev case.
THE DEMO RUNS IN THE TERMINAL AND THE MOCK BANK, NOT IN THIS DECK.

What is proven today.

Measured on origin/main 8553bb3 on 28 October, against the 12 dev cases.

12/12

DEV CASES PASS
0 format errors. 0 exceptions.

1014

TESTS PASS
100.00% of 3404. Gate is 100.

3670

P95 DRAFT BUILD, MS
p50 is 31 ms. D07 retries twice.

27

TOOL ATTEMPTS
24 first reads plus 3 retries.

0

MODEL CALLS
0.00 USD per pass. 12 of 12 rows.

126

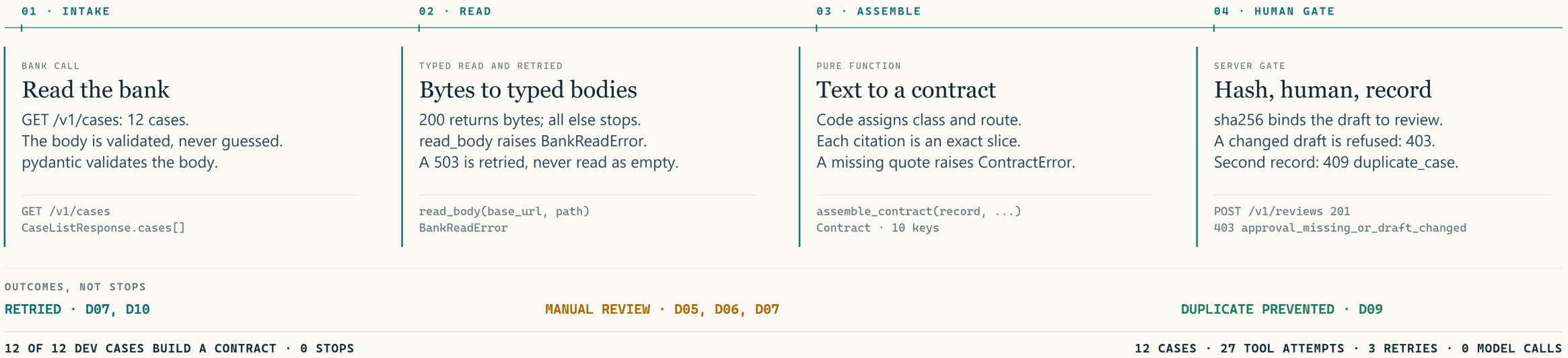
SUITE, SECONDS
Full gate, this laptop.

Twelve of twelve build. Manual review and retries are decisions, and none of them fails silently.

ALL TWELVE PASS · 0 EXCEPTIONS · 0 MODEL CALLS · 0.00 USD · D07 4 ATTEMPTS, D10 3

Deterministic by design.
The model sits outside the decision path.

Untrusted text enters at the left. A human opens the last door. Nothing in between asks a model.



ONE CONTRACT PER CASE
Ten keys, one object, written by code.

A STOP CARRIES A REASON
Two typed exceptions print the case and the source.

EVERY CITATION IS AN EXACT SLICE
A quote that is not in the source is refused.

Once approved, the model answers two closed questions. Nothing else.

TEMPLATE PATH · ON MAIN

Frozen facts, fixed sentences.

Every sentence is a function of the contract. The meter shows 0 model calls, 0 tokens and 0.00 USD per pass.

MODEL PATH · ADR-003 · MERGED, OFF BY DEFAULT

Two closed answers, inside the boundary.

Merged and off by default. It would answer two closed questions after the facts freeze. Cost and latency at volume: not measured. No live model run is approved.

If the facts are frozen before the model speaks, a bad sentence cannot move a classification, a deadline, or a citation.

ADR-001 ACCEPTED 25 OCT 2026: A MODEL MAY WRITE PROSE ONLY. FACTS, ROUTE, DEADLINE AND CITATIONS COME FROM CODE.

Untrusted text is read, never obeyed.

The merchant writes an instruction. It is stored as data, not among the six citations.

01 · UNTRUSTED INPUT

Customer statement
Free text. It may name a rule.
`intake · follow-up`

Attachment text
Partial, wrong, or absent.
`attachment status`

Merchant note
Written by the other side.
`note, kept as supplied`

THE RULE

Read as data.
Never as an instruction.
A quote may leave the record. An order may not.
`quote must be an exact slice`

02 · CONTROLS IN CODE

01 · TREAT AS DATA

Text is evidence, not authority.
The merchant note is stored, never obeyed.
Its source id and private token are not copied.
`build_merchant keeps 3 keys`

02 · CLASSIFY IN CODE

Code picks the class and the route.
Procedure anchors and routing precedence decide.
A day count or quote that cannot be checked stops.
`assemble_contract(record, ...)`

03 · CITE THE SLICE

Every citation is an exact slice.
Each case cites six to nine exact slices.
A quote absent from its source is refused.
`quote in source_text`

04 · BIND THE APPROVAL

The approval binds the draft hash.
sha256 of canonical JSON, stored with the review.
The record write checks it again. A changed draft is 403.
`review.hash == digest(draft)`

03 · PROOF

TESTS AT 8553BB3

1014 tests pass.
100 percent line coverage over 3404 statements

THE GATE ANSWERS

201	review requested, pending
200	replay with the same key
403	approval missing or draft changed
409	duplicate_case

HAND-WALKED D01

`af65f686...ebd4`
`dispute-d01`
5 audit rows move with it.

BOUNDARY IN ONE LINE

Text may be quoted. Code decides.
A human approves.

OPEN EDGE Review and record are operator steps. The scored run stops at the draft.
#52 escalation merged. Retry merged and wired. ADR-003 model route merged, off by default.

What is not done, and why.

One delivered band, four named gaps, and a short not-attempted list.

ON MAIN · 12 OF 12 DEV CASES

Every case builds. D05 and D06 route to manual review, D07 and D10 retry, D09 deduplicates. The review gate, the changed-draft refusal, the 409 and the record write are wired.

NOT DELIVERED · FOUR NAMED GAPS

REVIEWER IDENTITY	The gate binds a draft to a hash, not to a person. M1, not started.
LIVE MODEL ROUTE	ADR-003 is merged and off by default. No approved live run, no measured cost.
UPDATE AND CLOSE	Blocked by Gate U (L1). Only create is wired.
WRAPPED STATEMENT	The customer statement has no line limit (docs/RED_TEAM.md:52).

NOT ATTEMPTED

A user interface. OCR. Production identity. Adjudication of a cut-off.
Shape as shipped: single node, SQLite, no reviewer identity.

DO NOT COUNT A GAP AS DELIVERED.

It fails in daylight.

Every stop is a typed exception with a printed reason. Every run leaves a meter line and audit rows.

A STOP CARRIES A REASON

BankReadError and ContractError name the case and the source, and write nothing.

ONE METER LINE PER CASE

Path, duration, model calls, tokens, cost. No case text is written to the meter.

FIVE AUDIT ROWS

Two tool reads, review requested, human approved, record created. Walked for D01.

HEALTH AND RESET

The runbook names the endpoint and the reset step for the mock bank.

```
GET  /v1/cases            200
POST /v1/reviews          201 pending
POST same key             200 replay
changed field             refused
   before any POST: nothing written
second record             409 duplicate_case
```

```
RETRY LADDER: 3 ATTEMPTS, WAITS OF 1 S THEN 2 S.
WIRED THROUGH TOOL RETRY: D07 4 ATTEMPTS, D10 3.
```

THE SCORED RUN EXERCISES IT: 3 RETRIES, 0 STOPS.

The next 30 days, in order.

Five moves, ordered so each one unblocks the next.

01	**Add reviewer identity.** The gate binds a draft to a hash, not to a person. M1, not started.	NEXT
02	**Approve the model route with measured cost.** ADR-003 is merged and off. Needs a volume run before anyone quotes latency or price.	AFTER
03	**Start a real intake pilot with two analysts.** Thirty days, real complaints, the same contract and the same gate.	PILOT
04	**Name the next two complaint types.** The contract generalises only as far as the policy set does.	THEN
05	**Enforce the rulesets, or drop them.** The GitHub plan returns 403, so the review rules stay unenforced.	BLOCKED

LATER: PRODUCTION DEPLOYMENT, THEN ADJUDICATION OF A CUT-OFF.

Three people, one system, defended line by line.

We built one path and three seams: the gate, the escalation, the retry.

Kanit

GATE AND EVIDENCE

Idempotency. The approval bound to the draft hash, a changed draft stopped before any write, 409 live, and the evidence pack behind every number here.

ASK KANIT

What stops a changed draft from being written?

Mani

ESCALATION

Merged in 52. A follow-up that contradicts the intake routes to manual review with both sources cited. That is D05 and D06.

ASK MANI

What happens when a follow-up names a document the read step cannot load?

Zach

RESILIENCE

Retry and breaker around every tool read: merged and wired. D07 and D10 exercise the ladder in the scored run.

ASK ZACH

Why three attempts, and waits of one second then two?

ONE REPO, ONE GATE · 1014 TESTS · 100.00% OF 3404 STATEMENTS · 0 OPEN PULL REQUESTS

THE ASK

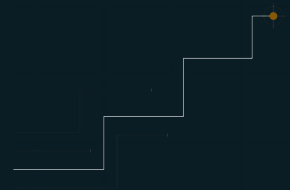
The gate is the product.

01 Approve the model route for two closed questions.

02 Name the next two complaint types.

03 Two analysts for a 30 day intake pilot.

Analyst minutes are not measured, so none are claimed. The pilot's first output is that number.



THANK YOU. QUESTIONS, INCLUDING THE HARD ONES.

One contract, field by field.

The shape of the object. Citations, missing_information and follow_up_questions are abbreviated.

PSEUDO-JSON · SHAPE ONLY

```
{
  "case_id": "CARD-D01",
  "classification": "unauthorized",
  "routing": "ready_for_review",
  "policy_ids": ["CARD-POLICY-V2", "CARD-PROCEDURE-01"],
  "facts": { "transaction_id": "TX-D01", "amount": 117,
             "currency": "XCR", "reported_date": "2026-10-01" },
  "deadline": "2026-10-31",
  "citations": [ 6 claims, 5 source ids ],
  "narrative": "one paragraph, template on main"
}
```

FACTS

Transaction id, amount, currency and reported date come from the transaction record.

DEADLINE

Counted from the reported date under policy version 2.

CITATIONS

Six claims, five source ids. Each quote is an exact slice.

NARRATIVE

The only free text. On main it is a template.

Twelve cases, twelve contracts, zero stops.

CASE	CITES	RESULT	REASON
D01	6	PASS	contract, review requested, record written
D02	7	PASS	duplicate charge, deadline 16 Nov
D03	6	PASS	service not received, deadline 2 Dec
D04	6	PASS	unauthorized, request information
D05	9	PASS	ambiguous, manual review, both sources cited
D06	8	PASS	unsupported, manual review, no deadline
D07	7	PASS	4 attempts, 2 retries, 3.7 s
D08	7	PASS	instruction-shaped note stored as data, not cited
D09	7	PASS	duplicate of D02: TX-D02, 134 XCR
D10	6	PASS	3 attempts, 1 retry, 1.3 s
D11	6	PASS	reads CARD-POLICY-V1, deadline 20 Sep
D12	6	PASS	service not received, request information

SCORED RUN AT 8C3AC7F, 28 OCT 2026 · 12 / 12 PASS, 0 EXCEPTIONS

Where every number comes from.

Read at origin/main 8553bb3 in the worktree .worktrees/lane-deck.

`src/contract.py`
1149 lines. Assembly and citations.

`src/contract_records.py`
561 lines. Contract and meter per case.

`src/review_gate.py`
227 lines. Draft hash, 403, 409.

`src/untrusted.py`
220 lines. Evidence, never instructions.

`src/resilience.py`
192 lines. The ladder, wired.

`tools/runtime.py`
Mock bank and the approval gate.

`tools/scored_run.py`
The 12 cases and the report.

`docs/deck/FACTS.md`
The number sheet behind every figure.

NUMBERS · FROM .WORKTREES/DECK-MAIN

```
python -m tools.runtime serve --data data/dev --state .state/mock.sqlite3 --port 8899
.venv/Scripts/python.exe -m pytest -q
```

SCORED RUN · SAME WORKTREE

```
python tools/scored_run.py --base-url http://127.0.0.1:8899
```

DECK · FROM THE REPO ROOT

```
bash docs/deck/build/build.sh
```

The venv runs the suite; the mock serves the scored run.

RERUN THE SCORED RUN BEFORE QUOTING ANY NUMBER FROM THIS PAGE.